

Metadata-Attentive Neural Recommendation with Feature-Level Attribution for Restricted Cold-Start Movie Recommendation

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Abstract—Neural Recommender Systems can model non-linear user-item interactions, but their latent representations offer little interpretability as to which item attributes are used to make the prediction. This limitation is more serious in item-cold-start settings, where collaborative interaction histories are sparse or missing. In this paper, we present an empirical study of metadata-aware neural recommendations for restricted cold-start movie recommendations. We propose a framework that combines collaborative user-item embeddings with structured movie metadata, such as genre indicators and Tag Genome semantic descriptors. The feature-level attention module assigns weights to metadata attributes, which can be viewed as attribution signals on the item side. Experiments on the MovieLens 20M dataset show that the proposed model reduces the warm-start RMSE from 1.0249 to 0.9629, and moderately improves ranking metrics over a collaborative-only neural baseline. We also evaluate the proposed model under a restricted metadata-assisted cold-start protocol, in which we remove some item interactions during training while keeping the metadata available, and the RMSE decreases from 1.1861 to 0.9628. The ablation study shows that the metadata contributes most to the performance gain, while the attention provides further improvement and inspectable feature weighting. Experimental results show that the proposed structured metadata attention can improve neural recommendation in our evaluated setting, and also serve as a light-weight way to investigate item-side metadata contributions.

Index Terms—Explainable Recommendation, Neural Collaborative Filtering, Attention Mechanism, Item Metadata, Cold-Start Recommendation, MovieLens, Tag Genome

I. INTRODUCTION

In this paper, we present an applied metadata-attentive neural recommendation framework that combines collaborative representation learning with feature-level attention over structured item metadata.

Explainable Recommendation methods attempt to provide deeper insight into the recommendation generation pro-

cess. Existing approaches include feature-based explanations, knowledge-aware methods, textual explanations, and attention-based mechanisms. However, many existing explanation techniques are applied after the model has generated predictions, resulting in post-hoc interpretations rather than explanations integrated directly into the learning process [1].

At the same time, many recommendation domains provide rich, structured metadata describing item characteristics. In movie recommendation, metadata such as genres and semantic tags provide human-understandable descriptions of item attributes. Such information is commonly used in hybrid recommendation systems to improve accuracy, but the explicit analysis of individual metadata feature contributions is rarely performed. Consequently, it remains unclear which semantic attributes contribute most strongly to a model’s recommendation decision [2].

Attention Mechanisms provide a potential solution by allowing models to learn different importance values for different input features. Instead of treating all metadata attributes equally, feature-level attention enables the model to assign adaptive importance weights to individual metadata dimensions during prediction. In this work, attention weights are interpreted conservatively as attribution signals representing learned feature importance rather than direct causal explanations of model behavior [3].

The item cold-start problem refers to the situation that newly introduced or rarely rated items do not have sufficient interaction histories to support reliable collaborative learning. To mitigate this limitation, recommendation models often utilize item metadata, such that the content attributes can compensate for the absence of the rating information. Such approaches have shown promising results, but the evaluation protocols used in prior work do not always clearly specify the metadata

and interaction information available at test time. This lack of transparency makes it difficult to compare methods and to understand how well metadata supports recommendations in cold-start settings [4].

In order to avoid unrealistic assumptions, we study a restricted metadata-assisted cold-start setting where the interactions of the selected item are hidden at training time, but structured item metadata is still available. In this paper, we propose an applied explainable neural recommendation framework that integrates collaborative representation learning and feature-level attention over structured item metadata. The proposed approach learns user and item representations by considering movie genres and semantic Tag Genome attributes through an attention-based metadata module. The resulting attention distribution enables us to analyze the contribution of metadata while maintaining competitive recommendation performance.

The aim of this work is to perform an empirical applied study of metadata-assisted neural recommendation in a constrained item cold-start setting. Rather than designing a very complex recommender architecture, we study the impact of a structured movie metadata on rating prediction, top-N ranking, and item side attribution in the case of limited collaborative item history. This framing is important because in practice Recommender Systems often have to work with the item descriptors that are available, e.g. genres and semantic tags, even when the user interaction history is sparse.

The main contributions of this work are as follows:

- We present an applied metadata-attentive neural recommendation framework that combines collaborative user-item embeddings with structured movie metadata derived from genres and Tag Genome descriptors.
- We introduce a feature-level metadata attention module that learns relative weights over metadata attributes and provides item-side attribution signals for inspecting model behavior.
- We define a restricted metadata-assisted item cold-start protocol in which selected item interactions are removed from training while metadata representations remain available.
- We evaluate the framework on MovieLens 20M using rating prediction metrics and top-N recommendation metrics, and compare collaborative-only, metadata-concatenation, and metadata-attention variants.
- We provide attention-based metadata attribution analysis and restricted cold-start case studies using human-readable genre and Tag Genome feature names.

The rest of the paper is organized as follows. Section II discusses related work on neural collaborative filtering, Metadata-Aware Recommendation, Attention Mechanisms, and Explainable Recommendation. Section III presents the proposed methodology. Section IV describes the dataset, preprocessing, experimental setup, and evaluation protocol. Section V reports rating prediction, ablation, convergence, and ranking results. Section VI presents the attention-based attribution analysis.

Section VII discusses limitations, and Section VIII concludes the paper.

II. RELATED WORK

A. Neural Collaborative Filtering

Collaborative filtering has been one of the most studied methods in Recommender Systems due to its ability to discover the latent user preference patterns from the user-item interaction data. Traditional matrix factorization methods represent users and items as low-dimensional latent vectors and use linear similarity functions to predict preferences. These approaches are effective but limited by their reliance on fixed interaction functions, such as the inner product, which may not capture complex relationships between users and items [5].

To overcome this limitation, He et al. proposed Neural Collaborative Filtering (NCF), which substitutes the hand-crafted similarity functions with neural networks to learn nonlinear interaction patterns between user and item embeddings. Their results showed that by adding linear matrix factorization components into multilayer neural networks, their NeuMF architecture outperforms traditional collaborative filtering methods in recommendation. Neural recommendation models have been paying more and more attention to learning richer representations of users and items. We exploit neural collaborative representation learning enriched with structured item metadata and feature-level attribution to provide interpretability.

B. Dual Representation Learning for Recommendation

Effective representation learning of both users and items is crucial for accurate recommendations. There are a few studies that have considered dual modeling strategies where the user and item representations are optimized separately to improve robustness under sparse interaction conditions.

DRMF (Dual-Regularized Matrix Factorization with Deep Neural Networks) proposed neural regularization approaches for latent representations of users and items. It showed that the independent modeling of the two sides of the recommendation process can improve the prediction performance under sparse data settings [6].

Similarly, Bayesian Dual Neural Networks studied different neural representation learning for users and items. Their work has demonstrated that dual modeling approaches can enhance recommendation quality, particularly in situations with low interaction information [7].

Drawing inspiration from these approaches, we utilize neural representation learning for users and items in our framework. However, we present an applied metadata attention mechanism to analyze the contribution of interpretable item attributes to the recommendations, rather than simply improving the quality of the latent representation.

C. Explainable Recommendation Systems

Neural Recommender Systems have demonstrated promising prediction accuracy, but they often make it hard to interpret their decision processes. Latent embeddings produced by deep

models are not directly interpretable, making it hard to explain why some items are recommended.

Zhang and Chen provided a comprehensive survey on Explainable Recommendation systems and emphasized the importance of transparency, trust, persuasiveness, and user satisfaction in the recommendation settings. Existing explainability methods include feature-based explanations, knowledge-based reasoning, textual explanations, and attention-based methods [?].

Existing methods are post-hoc analysis, i.e., they provide explanations after the recommendation generation. On the other hand, our method learns attention distributions over structured metadata features during prediction, thus directly incorporating interpretability into the recommendation architecture.

D. Attention-Based Recommendation Models

Attention Mechanisms have been widely used in recommendation models, enabling networks to weight different components of the input differently. Attention-based models can learn adaptive weights to identify the most influential information for recommendations.

Attentive Multi-Persona Collaborative Filtering (AMP) models multiple latent personas for different aspects of user preferences by an attention mechanism. The model shows that attention can enhance recommendation performance and provide interpretable preference information [8].

Other attention-based methods focus on sequential recommendation, where user interaction histories are modeled as ordered sequences. For example, to emphasize important past user behaviors while predicting future interactions, Attention Mechanisms have been combined with recurrent neural architectures.

Most existing approaches are either based on modeling user behavior sequences or latent preference representations, while our approach directly targets structured item metadata. The learned attention weights capture the relative importance of the movie genres and the semantic Tag Genome features for the recommendation predictions.

E. Metadata-Aware and Cold-Start Recommendation

A key limitation of collaborative filtering methods is that they need historical interaction data. Collaborative models struggle to learn reliable representations for new items with few or no interactions, a problem known as the item cold-start problem.

To tackle this challenge, Metadata-Aware Recommendation methods are proposed to leverage extra item information. Metadata representations have been shown to reduce dependency on item identifiers and improve recommendation robustness for sparse or new items, as demonstrated by recent works such as Metadata-aware Multi-task Transformer Recommendation (M2TRec)

Like RDNMF, Neural Matrix Factorization (NMF) leverages collaborative signals with textual item metadata and

shows that semantic content features can improve the quality of recommendations in the case of sparse interactions [9].

We use these techniques, enhanced with structured metadata from the MovieLens 20M dataset (e.g., genre information and Tag Genome semantic attributes). In this work, we focus on a limited metadata-assisted scenario where item interactions are not available during training and only item metadata is available during evaluation, as opposed to fully collaborative cold-start evaluations that rely heavily on interaction history.

F. Research Gap

Prior work has shown the effectiveness of neural collaborative filtering, Attention Mechanisms, and Metadata-Aware Recommendation. However, four limitations remain relevant to the present study:

- Neural recommendation models are primarily built to optimize the predictive performance but lack interpretability on how each item feature contributes to the predictions.
- Explanation is typically provided as a post-hoc method and not built into the recommendation architecture itself.
- Feature-level attribution does not adequately capture the effect of structured metadata features such as genres and semantic tags.
- In many instances, the assumptions made about the availability of item information at test time are not made explicit by cold-start evaluations.

In order to overcome these constraints, we suggest a neural recommendation system that provides interpretable metadata-assisted evaluation for cold-start by combining collaborative representation learning with feature-level focus on structured item information.

III. PROPOSED METHODOLOGY

This section presents the proposed explainable neural recommendation framework. The architecture is shown in Fig. 1. The proposed architecture consists of three components: collaborative user-item representation learning, feature-level metadata attention and rating prediction. The attention module takes in structured item metadata, and outputs attribution weights over genre and Tag Genome features.

A. Problem Formulation

Let

$$U = \{u_1, u_2, \dots, u_M\}$$

represent the set of users and

$$I = \{i_1, i_2, \dots, i_N\}$$

the set of items. User preferences are modeled as a rating matrix

$$R \in \mathbb{R}^{M \times N}$$

where the observed entries r_{ui} denotes the explicit rating that user u for item i .

Overall Architecture of the Proposed Explainable Metadata-Attentive Neural Recommendation Framework

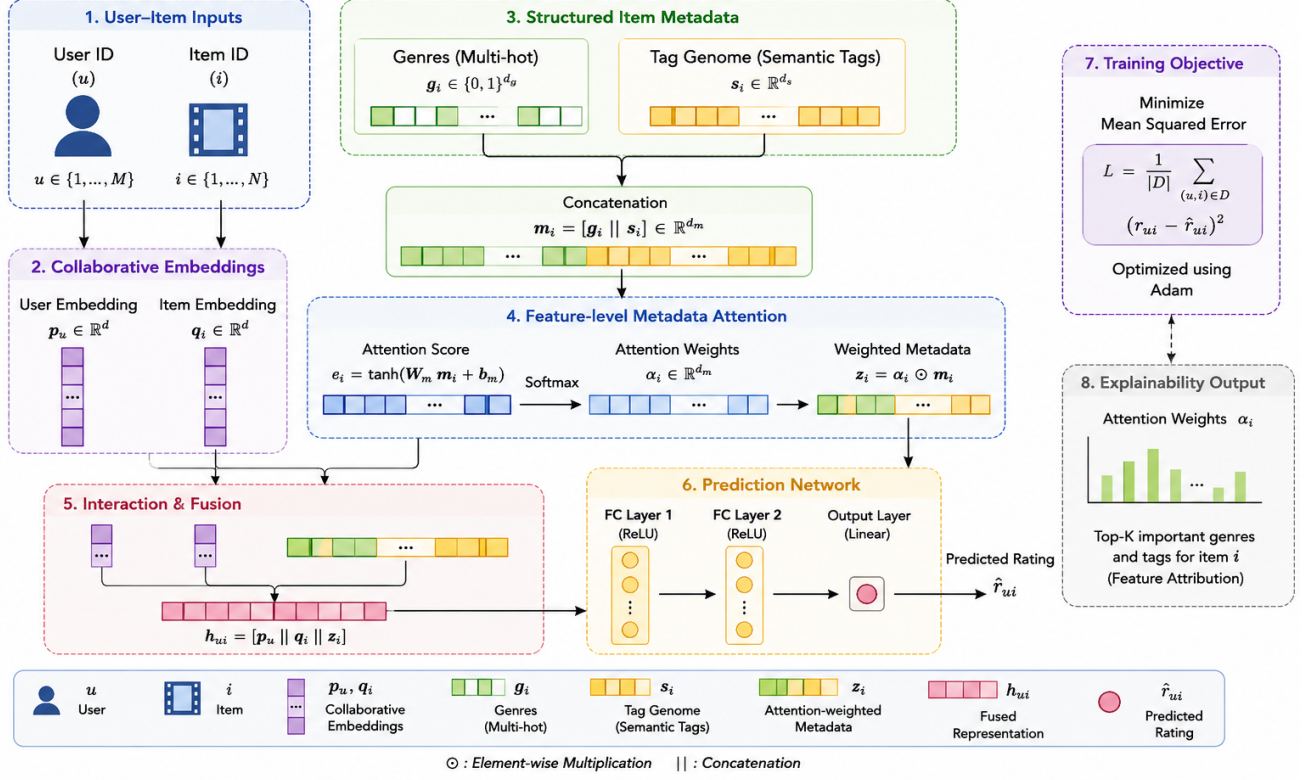


Fig. 1. Overall architecture of the proposed metadata-attentive neural recommendation model. User and item embeddings model collaborative signals, while genre and Tag Genome metadata are passed through a feature-level attention module. The resulting attention weights are used as item-side attribution signals and are not interpreted as causal explanations.

Given a user u , an item i , and its metadata representation m_i , The goal is to learn a prediction function:

$$\hat{r}_{ui} = f(u, i, m_i; \Theta)$$

where $\hat{r}_{ui}$ is the predicted rating and Θ are the trainable parameters of the model.

The attention module is item-conditioned because attention weights are computed from the metadata vector of the target item. Therefore, the extracted attention scores are interpreted as item-side metadata attribution rather than fully personalized causal explanations of user preference.

B. Neural Collaborative Representation Learning

The collaborative component learns latent representations for users and items using embedding layers. Each user id and item id is projected to a dense vector space.

The user embedding is defined as:

$$p_u = E_U(u)$$

and the embedding of the item is given by:

$$q_i = E_I(i)$$

where:

$$p_u, q_i \in \mathbb{R}^d$$

represent d -dimensional latent vectors, and E_U and E_I are learnable embedding matrices.

The learned embeddings are then fed into nonlinear neural layers, enabling the model to capture more complex user-item relationships, unlike traditional matrix factorization methods that are based on static inner-product interactions [10].

C. Structured Item Metadata Representation

Besides collaborative information, the proposed framework also includes structured metadata describing item characteristics. For each movie item, two types of metadata are considered.

First, genre information is represented as a multi-hot vector $g_i \in \{0, 1\}^{d_g}$, where each dimension indicates the presence or absence of a movie genre. Tag Genome descriptors are represented as a semantic relevance vector

$$s_i = [s_{i1}, s_{i2}, \dots, s_{ik}],$$

where s_{ij} denotes the relevance score of tag j for item i . The final metadata vector is obtained by concatenation:

$$m_i = [g_i \parallel s_i].$$

where each dimension represents presence or absence of a movie genre.

D. Feature-Level Metadata Attention

Standard hybrid recommendation models often just concatenate metadata features without explicitly modeling the importance of individual attributes [11]. However, different features of metadata may contribute unequally to recommendation decision.

To overcome this limitation, we present an applied feature-level attention mechanism on the metadata representation.

Given an item metadata vector $m_i \in \mathbb{R}^{d_m}$, the attention module first computes a feature-level score vector:

$$e_i = \tanh(W_m m_i + b_m), \quad e_i \in \mathbb{R}^{d_m}.$$

The normalized metadata attention distribution is computed as:

$$\alpha_i = \text{softmax}(e_i).$$

The attention-weighted metadata representation is then obtained as:

$$z_i = \alpha_i \odot m_i,$$

where $\odot$ denotes element-wise multiplication. The resulting vector z_i emphasizes metadata dimensions assigned higher attention by the trained model.

Because α_i is computed from m_i , the attention module is item-conditioned rather than user-conditioned. Therefore, the extracted weights indicate item-side metadata attribution and should not be interpreted as fully personalized explanations of user preference. The derived representation highlights metadata features that are paid more attention to and decreases the contribution of less important features.

E. Prediction Network

The collaborative and metadata-aware representations are concatenated as:

$$h_{ui} = [p_u \parallel q_i \parallel z_i].$$

The fused representation is passed through fully connected layers:

$$h_1 = \sigma(W_1 h_{ui} + b_1),$$

$$h_2 = \sigma(W_2 h_1 + b_2),$$

where σ denotes the nonlinear activation function. The final predicted rating is computed as:

$$\hat{r}_{ui} = W_o h_2 + b_o.$$

F. Training Objective

The model is trained by minimizing the mean squared error (MSE) over observed training interactions:

$$\mathcal{L} = \frac{1}{|D|} \sum_{(u,i) \in D} (r_{ui} - \hat{r}_{ui})^2,$$

where D denotes the set of observed user-item interactions, r_{ui} is the true rating, and $\hat{r}_{ui}$ is the predicted rating. Model parameters are optimized using the Adam optimizer.

G. Attention-Based Feature Attribution

After training, the learned attention weights are extracted to analyze the importance of metadata. For an item i , the highest-weighted metadata dimensions were selected as:

$$\text{TopK}(i) = \text{argsort}(\alpha_i, K)$$

These features correspond to the metadata dimensions receiving the highest attention-based attribution for the item. The attention weights are treated as attribution indicators and not as direct causal explanations [12]. Note that the attention weights are considered as feature attribution indicators and not as direct causal explanations.

H. Restricted Metadata-Assisted Cold-Start Evaluation

Let D_{train} denote the set of observed training interactions. For a selected cold-start item set $I_c \subset I$, all interactions involving these items are removed from training:

$$D_{\text{train}}^{\text{cold}} = \{(u, i, r_{ui}) \in D_{\text{train}} \mid i \notin I_c\}.$$

During evaluation, metadata m_i remains available for each $i \in I_c$. This setting evaluates whether structured metadata can support item representation when collaborative interaction history is unavailable during training.

IV. EXPERIMENTAL SETUP

This section describes the dataset, preprocessing routine, experimental settings and evaluation metrics for the proposed metadata-attentive neural recommendation framework.

A. Dataset

We test our approach on the MovieLens 20M benchmark dataset released by GroupLens. The dataset contains large-scale explicit user-movie rating interactions and structured movie metadata. Therefore, it is appropriate for evaluating the accuracy of recommendations and explainability based on metadata [13].

MovieLens 20M is a dataset with 20 million ratings from 138,493 users on 27,278 movies. Besides the ratings on interactions, the dataset also contains the movie genres and Tag Genome semantic information describing item-level characteristics.

The following components of the dataset are used:

- **Ratings:** Explicit user-movie feedback for collaborative representation learning

TABLE I
DATASET AND PREPROCESSING SUMMARY

Quantity	Value
Original Ratings	20,000,263
Original Users	138,493
Original Movies	27,278
Minimum Ratings per User	20
Minimum Ratings per Movie	50
Users After Filtering	8,000
Movies After Filtering	9,801
Ratings After Filtering	1,065,096
Genre Features	19
Tag Genome Features Used	100
Total Metadata Features	119
Metadata Normalization	Min-Max Scaling
Train/Test Split	80% / 20% (Chronological)
Validation Split	10% of Training Data
Cold-Start Items	10% of Items (980 Movies)
Random Seed	42

- **Genres:** Movie genres transformed into multi-hot features.
- **Tag Genome:** Descriptive attributes associated with movies represented by semantic relevance scores.

B. Data Preprocessing

A filtering approach was used in the preprocessing, which reduced data sparsity. The dataset was preprocessed by removing those users who had less than 20 ratings and those movies which had less than 50 ratings. We only reserve active users and movies with regular reviews, which has advantages of collecting more reliable interaction data and a more stable basis for the model training.

Each remaining user id and movie id are mapped to a continuous integer index for embedding-based learning. Movie metadata vectors are constructed by concatenating genre encodings with selected Tag Genome semantic descriptors [14].

The item metadata representation is finally:

$$m_i = [g_i || s_i]$$

where g_i denotes genre information and s_i denotes semantic Tag Genome features.

The 100 Tag Genome descriptors with the highest average relevance scores across movies were retained for metadata construction.

These processed data are then divided into training and testing sets for evaluation. Additionally, A restricted cold-start protocol was created by removing interactions associated with 10% of items from the training set while preserving their metadata representations.

C. Compared Models

The following model variants are evaluated:

- **Neural CF:** A collaborative-only neural recommender using user and item embeddings.
- **NCF + Metadata Concatenation:** A metadata-aware variant that directly concatenates user embeddings, item embeddings, and metadata vectors without attention.

TABLE II
MODEL HYPERPARAMETERS

Parameter	Value
User embedding dimension	64
Item embedding dimension	64
Metadata dimension	119
Hidden layers	128, 64
Activation function	ReLU
Optimizer	Adam
Learning rate	0.001
Loss function	MSE
Batch size	1024
Dropout	0.2
Epochs	10
Validation split	10% of training
Evaluation K	10
Random seed	42

- **Metadata Attention:** The proposed model applies feature-level attention over structured item metadata.
- **Cyclic Dual Model:** An additional representation-learning baseline that models user and item representations separately before prediction.

D. Implementation Details

All the models are implemented using a deep neural recommendation architecture and trained with the Adam optimizer. The embedding sizes for users and items are fixed at 64. The prediction network employs fully connected layers with nonlinear activation functions and dropout regularization [15].

The main hyperparameters are summarized in Table II.

E. Evaluation Metrics

We evaluate the models regarding rating prediction accuracy and top-N recommendation quality.

To predict the rating Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are used:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{j=1}^N (r_j - \hat{r}_j)^2}$$

$$\text{MAE} = \frac{1}{N} \sum_{j=1}^N |r_j - \hat{r}_j|$$

where r_j denotes the true rating, $\hat{r}_j$ denotes the predicted rating, and N is the total number of evaluated ratings. Precision@K measures the fraction of recommended items in the top (K) that are relevant. Recall@K measures the fraction of relevant test items recovered in the top (K). NDCG@K rewards relevant items appearing higher in the ranked recommendation list

We report ranking performance:

- Precision@K
- Recall@K
- Normalized Discounted Cumulative Gain (NDCG@K)

These metrics measure the presence of relevant movies in the top recommendations.

We also report the same metrics with the restricted metadata-assisted cold-start setting.

TABLE III
RESTRICTED COLD-START SPLIT SUMMARY

Quantity	Value
Cold-start item selection rule	Randomly select 10% of filtered items
Number of cold-start items	980
Interactions removed from training	All training interactions involving selected cold-start items
Metadata retained for cold-start items	Yes
Cold-start item embeddings used?	Yes, item IDs remain in the embedding matrix
Default/zero embedding used?	No
Embedding updates for cold-start items	No updates; no training interactions available
Evaluation users	Users appearing in <code>test_cold.csv</code>
Random seed	42

F. Restricted Metadata-Assisted Cold-Start Evaluation

To evaluate recommendation quality for items with limited collaborative information, we adopt a restricted cold-start protocol. After constructing the filtered MovieLens 20M dataset, ratings are chronologically divided into training (80%) and testing (20%) partitions. Subsequently, 10% of the training items are randomly selected and designated as cold-start items. All interactions associated with these items are removed from the training set, while their metadata representations remain available.

The resulting evaluation setting does not represent a fully zero-shot recommendation scenario. Instead, it examines whether structured metadata, including genres and Tag Genome descriptors, can provide meaningful item representations when collaborative interaction signals are unavailable. Cold-start evaluation is therefore based on metadata-assisted inference rather than historical user-item interactions.

Because cold-start item embeddings do not receive gradient updates, the cold-start evaluation mainly tests whether metadata-aware models can compensate for missing collaborative item history. The same cold-start split, random seed, and embedding initialization protocol are applied across all compared models [16].

For the top-N evaluation, we take the ratings of 4.0 and above as the relevant items. For each user that was tested, items in the warm test set are ranked according to their predicted ratings. For each user, the Precision@10, Recall@10, Hit Ratio@10 and NDCG@10 are computed and averaged over the evaluated users. We do not do any additional negative sampling. We use the same evaluation protocol and random seed across all the compared models.

V. RESULTS AND DISCUSSION

In this section, we provide the experimental evaluation of the proposed metadata-attentive neural recommender system. The evaluation is three-fold: accuracy of rating prediction, quality of top-N recommendation and behavior of training. The objective is to investigate the potential of using attention to

TABLE IV
RATING PREDICTION PERFORMANCE COMPARISON

Model	Warm Test		Cold Test	
	RMSE	MAE	RMSE	MAE
Neural CF	1.0249	0.8072	1.1861	0.9891
Metadata Attention	0.9629	0.7393	0.9628	0.7471

TABLE V
ABLATION STUDY COMPARING COLLABORATIVE-ONLY, METADATA-CONCATENATION, AND METADATA-ATTENTION MODELS.

Model	Warm RMSE	Warm MAE	Cold RMSE	Cold MAE
Neural CF	1.0249	0.8072	1.1861	0.9891
NCF + Metadata Concatenation	0.9653	0.7482	0.9705	0.7545
Metadata Attention	0.9629	0.7393	0.9628	0.7471

incorporate structured movie metadata in order to improve recommendation quality over collaborative-only representations.

A. Rating Prediction Performance

The warm-start RMSE is reduced from 1.0249 to 0.9629 and the MAE is reduced from 0.8072 to 0.7393, which are about 6.05% and 8.42% relative reductions, respectively, by the metadata-attention model. In the limited cold-start setting, the RMSE is reduced from 1.1861 to 0.9628 and the MAE is reduced from 0.9891 to 0.7471, which are relative reductions of about 18.83% and 24.47%. The larger cold-start improvement suggests that structured metadata is particularly useful when collaborative item information is not available at training time.

These results suggest that structured movie metadata provides stronger benefit when collaborative item information is limited. However, the reported improvements should be interpreted under the restricted metadata-assisted cold-start protocol rather than as a restricted metadata-assisted cold-start recommendation result [17].

B. Ablation Study

The ablation study separates the effect of adding metadata from the effect of attention-based metadata weighting. The NCF + Metadata Concatenation model improves substantially over Neural CF, showing that metadata provides useful item-side information. The Metadata Attention model gives a smaller additional improvement over direct concatenation, indicating that most of the gain comes from metadata availability, while attention contributes additional feature weighting and attribution [18].

The metadata-concatenation ablation is reported for rating prediction; ranking-level ablation is not included because the same ranking evaluation logs were not available for this variant.

C. Training Convergence Analysis

We study the training behavior of the examined neural recommendation models by observing the validation RMSE over epochs. Fig. 2 compares the convergence trends for the

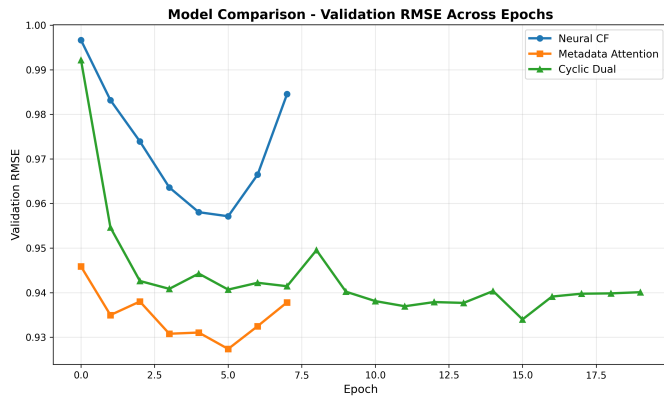


Fig. 2. Validation RMSE over the training epochs for the Neural CF, Metadata Attention and Cyclic Dual models. The Metadata Attention model has the lowest RMSE on the validation set, showing that the validation performance is improved by adding the movie metadata.

TABLE VI
TOP-N RECOMMENDATION PERFORMANCE COMPARISON

Model	Precision@10	Recall@10	HR@10	NDCG@10
Neural CF	0.6817	0.3045	0.9945	0.8571
Metadata Attention	0.6883	0.3123	0.9950	0.8620
Cyclic Dual	0.6830	0.3101	0.9939	0.8602

baseline, cyclic dual model, and proposed metadata-attention architecture.

The validation RMSE curves show that the metadata-attention model reaches lower validation error than the collaborative-only baseline during training. The baseline model begins to show mild overfitting after the middle epochs, while the attention-based model stabilizes earlier.

The cyclic dual model converges stably, but does not outperform the metadata-attention model. The convergence behaviour is compared using these curves and the final performance is separately reported on the test set.

D. Top-N Recommendation Performance

In addition to rating prediction, the recommendation ranking quality is evaluated by Precision@10, Recall@10, Hit Ratio@10 and NDCG@10. These metrics measure how well each model retrieves relevant movies and ranks them higher in the recommendation list.

The comparison results are given in Table VI.

Among these models, the Metadata Attention model obtains the best performance in terms of Precision@10, Recall@10 and NDCG@10. The improvements over Neural CF are small but consistent, indicating that metadata attention improves ranking quality along with rating prediction. The Hit Ratio@10 is almost saturated for all models, thus more informative ranking indicators such as Recall@10 and NDCG@10 are considered.

E. Discussion

Experimental results show that metadata is a useful complementary signal to neural recommendation. Unlike traditional

collaborative filtering methods that only consider historical interaction patterns, the proposed model explicitly models item features with the representations of genres and semantic tags.

The attention mechanism does not consider all the features equally but applies different importance weights to metadata attributes. These weights are feature-level attribution signals indicating which parts of the metadata contribute most to a model prediction.

It is worth highlighting that the interpretation of attention weights as model-based attribution indicators should not be confused with direct causal explanation. In the following section, we examine these attention distributions in more depth to analyze the explainability behavior of the proposed recommender system.

VI. EXPLAINABILITY ANALYSIS

Besides the predictive performance, we also study whether the metadata-attention module provides inspectable attribution over structured item features. Since the attention mechanism is computed based on the item's metadata, the analysis is more focused on attribution to the item side, rather than a fully personalized user-level explanation. The learned attention weights show which genre or Tag Genome dimensions the model weights more highly when constructing item representation.

Different from post-hoc explanation approaches, which assess the training model after prediction, the proposed method incorporates feature weighting into the recommendation architecture. The attention layer computes the relevance scores of the metadata embeddings based on the movie genres and semantic features from Tag Genome. The scores are feature-level attribution measures that signify which metadata aspects contribute more strongly to the learned recommendation representation [19].

The attention weights should be interpreted as attribution signals, and not as causal explanations. The higher attention value represents that the feature has a higher effect on the learned model representation, but does not represent that the feature is the causal factor of the recommendation decision.

A. Metadata Feature Attribution

Fig. 3 shows the most frequently occurring metadata attributes among the top-ranked attention features across cold-start movies. The figure shows the count of metadata attributes presence across the highest-attended features, not the raw attention weights.

The corrected attention analysis maps metadata indices to human-readable genre and Tag Genome feature names. This mapping is significant as numerical feature identifiers themselves are not meaningful explanations.

The attention distribution encodes the item-side metadata attributes that are emphasized in the representation construction. These scores are not causal explanations of a user's preference for a movie, but are treated as attribution indicators.

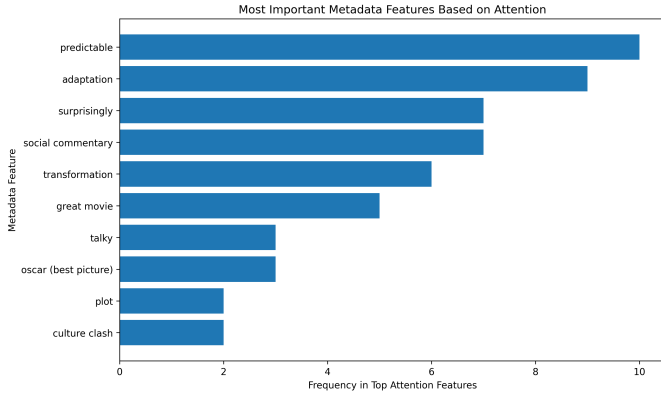


Fig. 3. Most frequently occurring metadata attributes among top-ranked attention features across cold-start movies. The plot reports feature frequency rather than raw attention weight; all feature indices are mapped to human-readable genre or Tag Genome names.

TABLE VII
RESTRICTED COLD-START CASE STUDIES USING
METADATA-BASED ATTRIBUTION.

Movie	Genres	Top Attention Features	Pred.
Stomp the Yard	Drama, Musical	Drama, Musical, adaptation	3.46
Pleasantville	Comedy, Drama, Fantasy	Comedy, Drama, Fantasy	3.54
Kingdom of Heaven	Action, Drama, War	Action, Drama, adaptation	3.65

B. Restricted Cold-Start Explanation

The case study illustrates how the model forms an item representation for a restricted cold-start movie using available metadata. Since the selected movies do not contribute interaction information during training, the explanation is derived primarily from genre and semantic Tag Genome descriptors rather than learned collaborative item history [20].

These examples show that the attention module is able to identify informative metadata features and provide interpretable explanations for model predictions in constrained cold-start settings. The explanations thus obtained should be viewed as a form of metadata based attribution rather than causal explanations of user preference.

These examples show that, when item interactions are removed from training, the model relies on available genre and semantic metadata to construct item representations. The table is intended as a qualitative inspection of metadata attribution rather than evidence of causal explanation.

C. Discussion on Interpretability

From the explainability analysis, we find two benefits of using attention over item metadata. It enables the recommender system to selectively emphasize important metadata rather than weighting all attributes uniformly. Second, it visualizes the learned significance scores as feature-level attribution signals. Compared to typical neural collaborative filtering models which produce recommendations based on latent representations that are difficult to interpret, the proposed framework

provides additional insight on what drives the recommendation results through attention-based metadata analysis. In the reported experiments, the added attribution mechanism does not reduce recommendation accuracy.

However, the generated explanations need to be used with care. Attention scores are a representation of the internal weighting patterns of the model and can give some insight into the usage of features, but they do not reveal causal correlations between metadata elements and user preferences.

VII. LIMITATIONS

Although the proposed metadata-attentive neural recommendation framework improves recommendation performance and provides metadata-level attribution, it still has several limitations. The attention mechanism provides feature-level attribution via learned metadata importance weights, but these weights should not be interpreted as direct causal explanations of user preferences. Attention distributions show how the model leverages available metadata at prediction time, but causality requires further analysis beyond the scope of this work [21].

Second, the cold-start evaluation in this work is a constrained metadata-assisted setting, rather than a full-fledged zero-shot recommendation scenario. The training data did not contain interactions with the selected cold-start items, but descriptive metadata were available. Thus, the results show the usefulness of structured metadata in situations where interaction information is scarce, but they do not determine how the system performs when there is no interaction history and no reliable metadata.

Third, the experimental evaluation is conducted on the MovieLens 20M dataset, which provides high-quality, structured metadata. In recommendation domains with incomplete/noisy/unavailable metadata, additional representation learning techniques or external knowledge sources may be needed [22].

Another limitation is that the current attention mechanism is item-conditioned and does not explicitly depend on the target user. Thus, the attribution analysis reveals metadata dimensions that are emphasized for item representation, but it does not yield fully personalized explanations of individual user preferences.

Finally, the model uses a light-weight feature-level attention mechanism and does not explore more complex user-conditioned, graph-based or multi-modal recommender architectures. Future extensions will explore richer semantic representations, multimodal metadata sources and sophisticated explanation validation techniques to further improve both recommendation accuracy and interpretability.

VIII. CONCLUSION

In this paper, an applied study of metadata-attentive neural recommendation for restricted cold-start movie recommendation was proposed. The proposed model combines collaborative user-item embeddings and structured movie metadata from genres and Tag Genome descriptors. The feature-level

attention module outputs weights for the dimensions of the metadata, which makes the item side attribution analysis possible while it is lightweight. Experiments on MovieLens dataset 20M show that metadata helps improve the rating prediction and ranking performance over a collaborative-only neural baseline. The ablation study shows that the metadata contributes the most to the performance gain, while the attention provides an extra gain and inspectable feature weighting. In the restricted metadata assisted cold-start protocol, the model outperforms the collaborative-only baseline, demonstrating that structured metadata is useful when item interaction history is not available at training time. Note that attention scores should be considered as attribution signals, not causal explanations. Future work involves user conditioned attention . better baselines for recommendation and explanation validation on larger recommendation datasets .

AVAILABILITY OF CODE AND DATA

To support reproducibility, the source code, preprocessing scripts, trained models, and evaluation procedures used in this study are publicly available at <https://github.com/Upkarkaur02/Metadata-Attention-Recommender>.

The experiments were conducted using the MovieLens 20M dataset and Tag Genome metadata, available from GroupLens at <https://grouplens.org/datasets/movielens/20m/>.

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